

Public earnings ranks: a reproducible association with strict disclosure limits

A standard education/age-proxy model remains competitive with more elaborate alternatives on reserved respondents. The strongest result is a well-scoped reproduction, not a new skill-return or absolute-wage law.

Data and evaluation

PIAAC USA/Japan public files suppress exact age and absolute/PPP hourly wages. Valid earnings deciles and source education among ages 25-65 yield 4236 development and 1054 confirmation respondents. Country-linked hash blocks preserve each person, with final SPFWT0 within blocks and equal country/block loss. All ten background-conditioned plausible values are excluded from predictors.

Target: **Upper three country-specific hourly-earnings deciles, excluding bonuses**, measured as binary $EARNHRDCLC2 \geq 8$. Primary error: squared probability (Brier). Contemporaneous cross-sectional earnings association using age/education/country. All 10 plausible numeracy values are excluded: conditioned on background and unsuitable individual prediction scores. This is not future earnings or causal education return.

Respondents nested in country and deterministic ten hash blocks; not 10 independent samples from plausible values. Repeated respondent information stays linked. Survey clustering not publicly fully reconstructible. Respondent hash blocks within both countries; no held respondent wage fitting. Country intercepts use training respondents in that same country. No transfer to an unseen economy.

Findings and equations

$$p = \text{logistic}(a + b_J J + b_S S + b_X X + b_{XX} X^2)$$

The notation and all fitted coefficients are defined below. Here logistic denotes the inverse-logit function; a positive-part subscript denotes $\max(x, 0)$.

P100-039-F01 - reproduction (partially_supported). A standard education and coarse experience-proxy association predicts upper within-country earnings ranks better than country prevalence and comparably to flexible controls.

$$p = \text{logistic}(a + b_J J + b_S S + b_X X + b_{XX} X^2).$$

Consistent with established Mincer-style labor associations, but the binary rank target is not a log-wage equation and coefficients are not percentage earnings returns.

Falsifying evidence and limits: Coarse age, unobserved actual experience, employment selection and country-specific ranks prevent causality/absolute earnings claims; numerical gain over flexible is modest.

P100-039-F02 - informative_falsification (supported). Absolute PPP-wage and independent individual skill-return hypotheses are not evaluable using these released fields.

No admitted absolute-wage/individual-PV equation; rejected before fitting.

Exact AGE_R and PPP earnings are all suppressed; OECD PVs are conditioned population-inference quantities, not individual scores.

Falsifying evidence and limits: SUPPRESSION_AUDIT.json and primary OECD methodology document the rejected prerequisites. New data/independent measurement would be required.

Selected executable model: domain. $S = \text{education years} - 12$; $X = \max(\text{age-bin midpoint} - \text{education years} - 6, 0)/10$; $J = \text{Japan indicator}$. $p(\text{top three country earnings deciles}) = \text{logistic}(a + b_J J + b_S S + b_X X + b_{XX} X^2)$. Coefficients in that order = [-3.1358399163295814, 0.11186246189215215, 0.370679096774003,

1.2373931631741675, -0.19547907390628247]. Midpoints encode published age bins, not exact age or work tenure; earnings ranks are country relative, not comparable dollar amounts.

All coefficients, input definitions, training groups and transformations are in `rules.json`; the runnable reference performs no fitting.

Compact numerical evidence

Errors are computed within declared complete groups (with final survey weights when supplied), then averaged equally across groups.

Frozen model	Development error	Confirmation error	Worst confirmation group
reference	0.191515	0.1820236	0.2103406
constant	0.2249295	0.2194144	0.2411918
domain	0.191515	0.1820236	0.2103406
flexible	0.192086	0.1836466	0.2108711

Confirmation contains 1054 scored observations in 4 groups. Individual-group scores and every attempted model remain available. Confirmation Brier 0.182024 for the development-selected domain model versus country prevalence 0.219414 and flexible 0.183647. More elaborate attempts range lower retrospectively, but all were within 1 percent of the best development model and the simpler domain model was frozen. Four held hash blocks are computational partitions, not four independent economies.

Interpretation, limitations and use

Only valid reported earnings deciles, ages 25-65 and complete education. Final weights within equal country×respondent blocks; not a pooled national population estimand. Potential experience age-schooling-6 is proxy, not measured tenure. No skill-wage causal claim; income reporting/topcoding and nonresponse remain. Whole-group errors/ranges; no row bootstrap or manufactured independent-population confidence. Confirmation becomes exposed after freeze.

Useful for benchmark discipline: suppressed monetary values and population-oriented plausible values cannot become invented individual targets/predictors. The executable task supports legitimate country-relative rank-association comparisons without claiming causal returns or individual skill measurement.

Reproducibility and prior art

Run `python run.py` to verify hashes and replay every frozen equation. Predictions use declared input columns only. The shared benchmark evaluator can score alternative equations under the same task; numerical agreement with these coefficients is not required. Public source-aware confirmation is now exposed. No independent experimental replication or certified novelty is claimed.

- Plausible values conditioned on background and intended for population inference, not individual scores. (checked 2026-10-02).
- Do not average 10 PVs into an individual score; final and replicate weights serve survey uncertainty. (checked 2026-10-02).
- Mincer-style earnings/skill associations are extensive prior art; no new discovery from recovering education/experience gradients. (checked 2026-10-02).